MODELLING WOMEN'S LABOUR FORCE PARTICIPATION

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STAT 311

Contents of Contents

1. Introduction 1
2. Exploratory Data Analysis 2
3. The Probit Model 3
4. Post Hoc Analysis 4
5. Key Findings 5
6. Recommendations 5
7. References 5
8. Appendix 6
   - 8.1. Summary statistics 6
   - 8.2. EDA Tables 6
     - 8.2.1. Participation by children under six 6
     - 8.2.2. Participation by the years of education 6
   - 8.3. Null Probit Model Summary 7

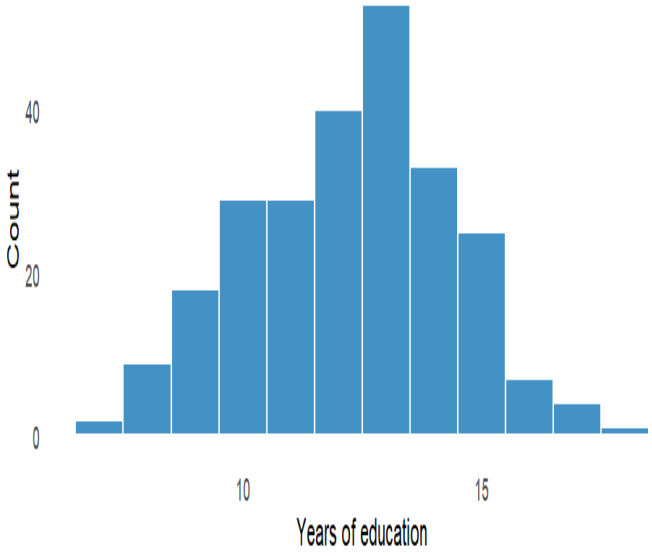
1. Introduction

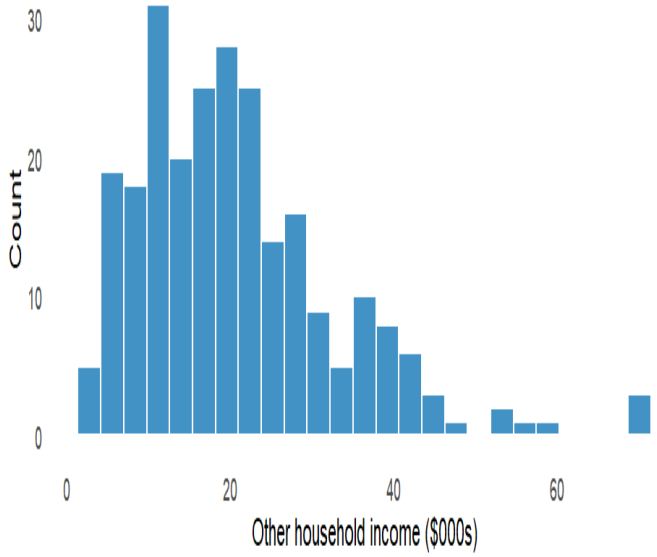
This report examines the determinants of women's labour force participation using a probit model. The objective is to estimate the probability that a woman participates in the labour force based on three characteristics: years of education, the number of children under the age of six in the household, and other household income. The analysis is based on a sample of 250 women, Of these, 114 women (45.6%) were in the labour force at the time of observation and 136 women (54.4%) were not.

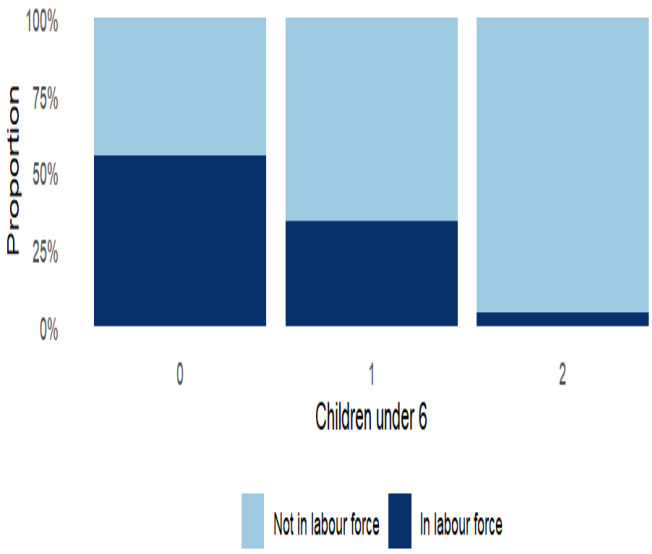
Ordinary Least Square does not work with Yes/No choices like labour force participation, because it can predict impossible probabilities like below 0 or above 1 which breaks basic statistical rules. A probit model solves this problem by assuming there is an invisible underlying score(for example inner drive or personal readiness to work), this keeps all predicted probabilities neatly between 0% and 100%.

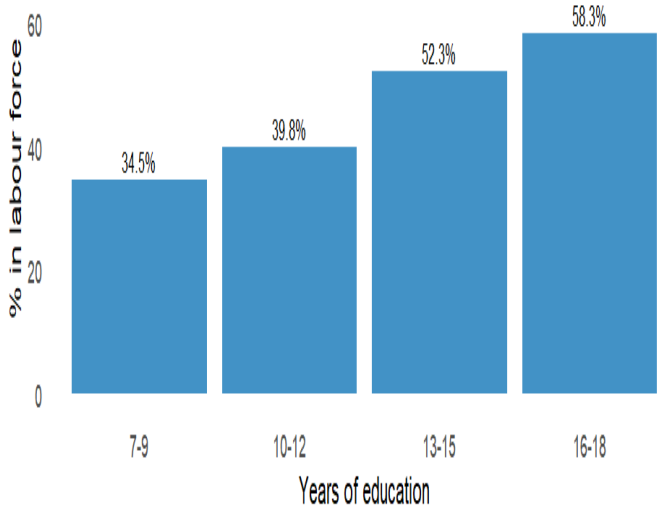
This analysis proceeds in 4 stages; first presents the EDA, then estimates the probit model, thirdly we'll undertake a post hoc analysis and finally discuss key findings and offer recommendations.

2. Exploratory Data Analysis









3. The Probit Model

Coefficients	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	-0.945228	0.516358	-1.831	0.067165 .	
educ	0.139499	0.041700	3.345	0.000822	***
kidslt6	-0.753319	0.154610	-4.872	1.1e-06	***
nwifeinc	-0.030200	0.007823	-3.861	0.000113	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The fitted model is:

$$P(inlf=1|X) = \Phi(-0.9452 + 0.1395education - 0.7533kidslt6 - 0.0302nwifeinc)$$

All three predictors are significant, the null deviance is greater than the residual deviance, The reduction of 52.51 across the three predictors indicates the model explains a meaningful share of the variation in participation, compared with an intercept-only model.

According to the above model, each additional year of education raises the latent propensity to participate by 0.1395 standard deviations, holding the other predictors constant. Which is Consistent with the education-band table, as seen here where participation rose from 34.5% to 58.3% across the bands. On the other hand, each additional child under six lowers a woman's latent propensity to participate by 0.7533 standard deviations. This is the largest coefficient in the model, matching the bivariate finding where participation fell from 55% to 4.3% as seen here. Lastly, each additional $1,000 of other household income lowers the latent propensity by 0.0302 standard deviations.

To enhance understanding of the probit model coefficient interpretation, we'll look at an example whereby a woman has 12 years of education, no children under 6, and $20,000 other household income which is given by;

$Z= -0.9452 + 0.1395(12) - 0.7533(0) - 0.0302(20)= 0.1248$

Therefore, her predicted probability of participating would be roughly 55%.

Take the same woman but this time with 1 child under 6 years ;

$Z= -0.9452 + 0.1395(12) - 0.7533(1) - 0.0302(20)= -0.6285;\ \Phi(-0.6285)= 0.265$

Adding a single child under six to this profile reduces the predicted probability of labour force participation from 55.0% to 26.5% ; a fall of 28.5 percentage. This is consistent with the with the bivariate analysis we did as seen here, and this illustrates why the raw coefficient (−0.7533) is not itself the answer as the same coefficient produces different changes in probability depending on where on the curve the woman sits.

4. Post Hoc Analysis

Marginal effects at the reference case

Reference woman: 12 years of education, no children under 6, $20,000 other household income → baseline probability 0.5496 (55.0%)

Change	Effect on probability
+1 year of education	+5.5 points (0.5496 → 0.6042)
+1 child under 6	−28.5 points (0.5496 → 0.2648)
+$10,000 other household income	−12.0 points (0.5496 → 0.4297)

One child under six moves the probability five times more than a year of education and roughly two-and-a-half times more than a $10,000 income difference.

Because the probit model is nonlinear in the predictors, the effect of a one-unit change depends on where the individual sits on the curve. Marginal effects were therefore computed at a reference case; a woman with 12 years of education, no children under six, and $20,000 in other household income .

Education	No child under 6	One child under 6	Change
9 years	0.3845	0.1475	−0.2369
12 years	0.5496	0.2648	−0.2848
15 years	0.7065	0.4168	−0.2897
18 years	0.8319	0.5826	−0.2494

The same coefficient produces different changes in probability depending on the starting point. Adding a child costs 23.7 points for a woman with 9 years of education, peaks at 29.0 points at 15 years, and falls to 24.9 points at 18 years. It's largest in the middle of the distribution where the S-curve is steepest and smaller at both ends.

5. Key Findings

- All three predictors are statistically significant, years of education raise the probability of labour force participation; the number of children under six and other household income both reduce it.
- The dominant factor is the presence of young children. At the reference case, adding a single child under six lowers the predicted probability of participation by 28.5%; from 55.0% to 26.5% that is five times the effect of an additional year of education. In the bivariate analysis, participation fell from 55.0% among women with no young children to 4.3% among those with two.
- Education operates in the opposite direction, with participation rising from 34.5% in the lowest education band to 58.3% in the highest, and an additional year of education adding 5.5% to the predicted probability at the reference case.
- Other household income has a moderate negative effect, with a $10,000 difference reducing predicted participation by 12 %

6. Recommendations

- Given the marginal effect of the childcare relative to the other predictors, the most direct solution would be access to affordable childcare for children under six years.
- The model implies that policies reducing the effective burden of young children such as subsidised childcare, expanded spaces, or flexible arrangements would produce larger changes in participation than interventions targeting education or income alone

7. References

Baker, M., Gruber, J., & Milligan, K. (2008). Universal child care, maternal labor supply, and family well-being. *Journal of Political Economy, 116*(4), 709–745.

Baker, M., Gruber, J., & Milligan, K. (2026). *Investing in mothers? The long-run impact of a universal child care program on maternal work and income* (NBER Working Paper No. 35514). National Bureau of Economic Research.

Lefebvre, P., & Merrigan, P. (2008). Childcare policy and the labor supply of mothers with young children: A natural experiment from Canada. *Journal of Labor Economics, 26*(3), 519–548.

8. Appendix

8.1. Summary statistics

```
summary(labour)
     inlf             educ          kidslt6          nwifeinc
Min.   :0.000   Min.   : 7.00   Min.   :0.000   Min.   : 2.80
1st Qu.:0.000   1st Qu.:11.00   1st Qu.:0.000   1st Qu.:11.53
Median :0.000   Median :12.00   Median :0.000   Median :18.95
Mean   :0.456   Mean   :12.24   Mean   :0.408   Mean   :20.88
3rd Qu.:1.000   3rd Qu.:14.00   3rd Qu.:1.000   3rd Qu.:27.20
Max.   :1.000   Max.   :18.00   Max.   :2.000   Max.   :70.10
```

8.2. EDA Tables

8.2.1. Participation by children under six

Children under 6	n	Participated	Rate
0	171	94	55.0%
1	56	19	33.9%
2	23	1	4.3%

8.2.2. Participation by the years of education

Education	Rate
7–9 years	34.5%
10–12 years	39.8%
13–15 years	52.3%
16–18 years	58.3%

8.3. Null Probit Model Summary

```
Call:
glm(formula = inlf ~ 1, family = binomial(link = "probit"), data = labour)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.11052    0.07944  -1.391    0.164

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 344.64  on 249  degrees of freedom
Residual deviance: 344.64  on 249  degrees of freedom
AIC: 346.64

Number of Fisher Scoring iterations: 3
```

8.4. Full Probit Model Summary

```
Call:
glm(formula = inlf ~ educ + kidslt6 + nwifeinc, family = binomial(link = "probit"),
    data = labour)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.945228   0.516358  -1.831 0.067165 .
educ         0.139499   0.041700   3.345 0.000822 ***
kidslt6     -0.753319   0.154610  -4.872  1.1e-06 ***
nwifeinc    -0.030200   0.007823  -3.861 0.000113 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 344.64  on 249  degrees of freedom
Residual deviance: 292.13  on 246  degrees of freedom
AIC: 300.13

Number of Fisher Scoring iterations: 5
```

8.5. R Code

```
# ---- Correlation matrix ----
round(cor(labour), 3)


library(ggplot2); library(gridExtra); library(dplyr)


p1 <- ggplot(labour, aes(x = educ)) +
  geom_histogram(binwidth = 1, fill = "#4292c6", colour = "white") +
  labs(title = " ", x = "Years of education", y = "Count") +
  theme_minimal() + theme(panel.grid = element_blank(), plot.margin = margin(t=5,r=5,b=2,l=2))


p2 <- ggplot(labour, aes(x = nwifeinc)) +
  geom_histogram(bins = 25, fill = "#4292c6", colour = "white") +
  labs(title = " ", x = "Other household income ($000s)", y = "Count") +
  theme_minimal() + theme(panel.grid = element_blank(), plot.margin = margin(t=5,r=5,b=2,l=2))


p3 <- ggplot(labour, aes(x = factor(kidslt6), fill = factor(inlf))) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("0"="#9ecae1","1"="#08306b"),
                    labels = c("Not in labour force","In labour force"), name = NULL) +
  scale_y_continuous(labels = scales::percent) +
  labs(title = " ", x = "Children under 6", y = "Proportion") +
  theme_minimal() + theme(panel.grid = element_blank(), legend.position = "bottom",
                          plot.margin = margin(t=5,r=5,b=2,l=2))
p4_data <- labour %>%
  mutate(educ_band = cut(educ, breaks = c(6,9,12,15,18),
                         labels = c("7-9","10-12","13-15","16-18"))) %>%
  group_by(educ_band) %>%
```

```
  summarise(rate = mean(inlf)*100, .groups = "drop")

p4 <- ggplot(p4_data, aes(x = educ_band, y = rate)) +
  geom_col(fill = "#4292c6") +
  geom_text(aes(label = paste0(round(rate,1), "%")), vjust = -0.4, size = 3) +
  labs(title = " ", x = "Years of education", y = "% in labour force") +
  theme_minimal() +
  theme(panel.grid = element_blank(),
        plot.margin = margin(t=5, r=5, b=2, l=2)) +
  ylim(0, max(p4_data$rate) * 1.2)

labour %>% group_by(kidslt6) %>%
  summarise(n = n(), participated = sum(inlf),
            rate = round(100*mean(inlf),1), .groups = "drop") %>% print()

p4_data %>% print()

# ---- all four on one page ----
grid.arrange(p1, p2, p3, p4, nrow = 2, ncol = 2)

# ============================================================
# Probit model — Labour force participation
# ============================================================
probit <- glm(inlf ~ educ + kidslt6 + nwifeinc,
              family = binomial(link = "probit"),
              data = labour)

summary(probit)
```

```
# null model, for the null vs residual deviance comparison
probit_null <- glm(inlf ~ 1, family = binomial(link = "probit"), data = labour)
summary(probit_null)

# ------------------------------------------------------------
# 5. WORKED EXAMPLES — the two-step procedure
#    Step 1: compute z.  Step 2: look up Phi(z) with pnorm()
# ------------------------------------------------------------

# woman with 12 yrs education, no young children, $20k other income
z1 <- -0.9452 + 0.1395*12 - 0.7533*0 - 0.0302*20
pnorm(z1)

# same woman, but with one child under 6
z2 <- -0.9452 + 0.1395*12 - 0.7533*1 - 0.0302*20
pnorm(z2)

# the class's predict() approach
predictions <- predict(probit,
                       newdata = data.frame(educ    = c(12, 12),
                                            kidslt6 = c(0, 1),
                                            nwifeinc = c(20, 20)),
                       type = "response")
predictions
diff(predictions)



# ------------------------------------------------------------
```

```
# 7. PROBIT CURVE
#    (note the -1 on the outcome: factor -> numeric 0/1)
# ------------------------------------------------------------
ggplot(labour, aes(x = educ, y = as.numeric(inlf) - 1)) +
  geom_point(alpha = 0.25, colour = "#4292c6") +
  stat_smooth(method = "glm", se = FALSE, colour = "#08306b",
              method.args = list(family = binomial(link = "probit"))) +
  labs(title = " ", x = "Years of education", y = "In labour force") +
  theme_minimal() +
  theme(panel.grid = element_blank(),
        plot.margin = margin(t=5, r=5, b=2, l=2))

#Post Hoc Analysis
1. Marginal effect at a reference case
#    Reference woman: educ = 12, kidslt6 = 0, nwifeinc = 20

ref <- data.frame(educ = 12, kidslt6 = 0, nwifeinc = 20)

# baseline probability
predict(probit, newdata = ref, type = "response")

# effect of one additional year of education
predict(probit, newdata = data.frame(educ = 13, kidslt6 = 0, nwifeinc = 20),
        type = "response") - predict(probit, newdata = ref, type = "response")

# effect of one additional child under 6
predict(probit, newdata = data.frame(educ = 12, kidslt6 = 1, nwifeinc = 20),
        type = "response") - predict(probit, newdata = ref, type = "response")
```

```
# effect of $10,000 more other household income
predict(probit, newdata = data.frame(educ = 12, kidslt6 = 0, nwifeinc = 30),
       type = "response") - predict(probit, newdata = ref, type = "response")
2. How the effect of children varies across education levels
#    (the probit curve is non-linear, so effects differ by starting point)

educ_seq <- c(9, 12, 15, 18)

res_kids <- sapply(educ_seq, function(e) {
  p0 <- predict(probit, newdata = data.frame(educ = e, kidslt6 = 0, nwifeinc = 20), type = "response")
  p1 <- predict(probit, newdata = data.frame(educ = e, kidslt6 = 1, nwifeinc = 20), type = "response")
  c(no_child = p0, one_child = p1, change = p1 - p0)
})
round(t(res_kids), 4)

# ------------------------------------------------------------
# 3. Predicted probability across the education range
# ------------------------------------------------------------
educ_range <- data.frame(educ = 7:18, kidslt6 = 0, nwifeinc = 20)
educ_range$prob <- predict(probit, newdata = educ_range, type = "response")
round(educ_range, 4)
4. Predicted probability across the income range
# ------------------------------------------------------------
inc_range <- data.frame(educ = 12, kidslt6 = 0, nwifeinc = seq(5, 65, 10))
inc_range$prob <- predict(probit, newdata = inc_range, type = "response")
round(inc_range, 4)
```